

SpatialLLM: A Multi-Agent Framework for Adaptive and User-Aware Mixed Reality Content Placement

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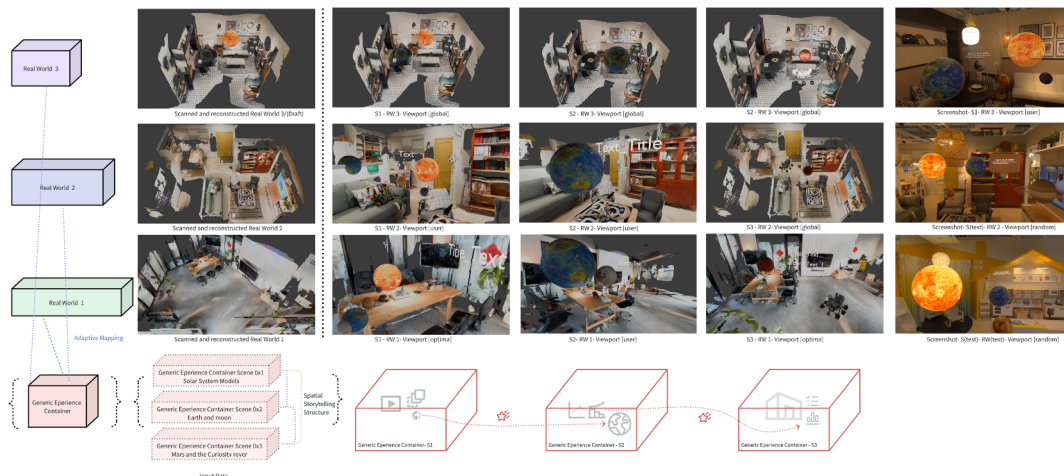


Figure 1: Example results across three real-world indoor environments. For each environment, we show the SLAM-based 3D reconstruction and representative placements produced by SpatialLLM. The system adapts layout decisions to local geometry (e.g., support surfaces and free space) while maintaining scene-level coherence across rooms.

ABSTRACT

Presenting information in Mixed Reality is essential for immersive applications, yet current systems largely depend on static 3D layouts or manual placement, which limits adaptability across diverse environmental contexts. We introduce **SpatialLLM**, a multi-agent framework leveraging Large Language Models (LLMs) to place adaptive, scene-aware MR content. SpatialLLM grounds LLM intent reasoning in a constraint-verified spatial memory (ST-CM) that enforces physical feasibility (support contact, collision avoidance, free space) and scene continuity via constrained re-generation. Across three real-world indoor environments, average placement error drops from 65.2% (single-prompt baseline) to 8.6%. A user study (n=11) reports improved coherence, stability, and adaptability.

Index Terms: Mixed Reality, Large Language Models, Multi-Agent Framework, Adaptive Content Placement, Spatial Reasoning.

1 INTRODUCTION

Mixed Reality (MR) applications often anchor virtual objects to a specific physical space. When users move across rooms or contexts, placements can become misaligned, collide with geometry, or violate everyday spatial expectations, reducing coherence and usability.

Recent work integrates large language models (LLMs) into MR to improve contextual understanding and interaction, such as LLM-assisted analytics for understanding user behaviors in XR [3], and

context-aware assistance grounded in real-world sensing [4]. However, most LLM+XR systems emphasize instruction or assistance rather than *executable* layout synthesis that remains physically feasible under sensed geometry and coherent across scenes [2]. This gap is particularly salient when layouts must be adapted online under sensed geometry and semantic constraints [1].

We introduce **SpatialLLM**, a multi-agent framework for adaptive, user-aware MR content placement. SpatialLLM separates semantic reasoning from geometric verification: agents infer intent and propose constraints, while a constraint-verified spatial memory (ST-CM) validates feasibility and scene continuity before executing placements.

Contributions. (1) A multi-agent MR placement pipeline for intent understanding, constraint derivation, and executable layout synthesis. (2) A constraint-verified spatial memory (ST-CM) that enforces feasibility (support, collision-free placement) and scene continuity. (3) Evaluation across three environments (65.2% $\rightarrow$ 8.6% placement error) and a user study (n=11) showing improved perceived coherence and adaptability.

2 FRAMEWORK

SpatialLLM is a coordinated system of specialized LLM agents and supporting modules (Figure 2). We define adaptive MR content placement over three inputs: (1) physical environment data from an MR device (e.g., detected planes/anchors and reconstruction), (2) a project specification describing intended content and scene flow (Generic Experience Containers), and (3) an interaction toolkit exposing high-level APIs for object manipulation and behaviors. The system outputs an executable MR layout that is both intent-aligned and physically feasible.

Spatial Memory and Constraint Layer (ST-CM). To make LLM-generated spatial decisions executable and robust, SpatialLLM in-

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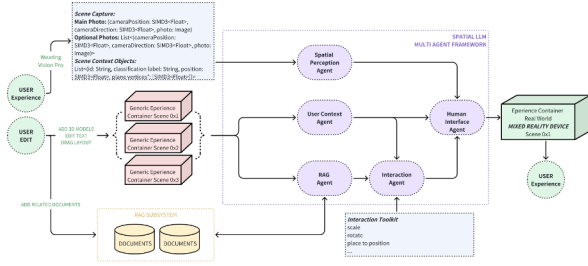


Figure 2: SpatialLLM overview. Agents propose intent-aligned constraints and placements; the spatial memory and constraint layer (ST-CM) validates feasibility and continuity before emitting executable layout commands.

stantiates a *Spatial–Temporal Cognitive Map (ST-CM)* that maintains (i) a structured representation of sensed geometry (planes, anchors, free-space regions) and (ii) scene-level object states and relations across time. Agents produce proposed constraints and candidate placements, which ST-CM verifies using feasibility rules (support contact, non-penetration/collision, and reachable placement regions). Violations trigger constrained re-generation or correction, reducing brittleness under LLM non-determinism.

Grounding and I/O. LLMs do not consume raw camera pose streams directly. The Spatial Perception Agent converts device sensing into a compact scene summary (e.g., room label, planes, anchors) stored in ST-CM and serialized for agent reasoning. Agents then emit machine-readable candidates (object $\rightarrow$ surface/region $\rightarrow$ pose $\rightarrow$ anchor type), e.g., `spawn(object, plane_id, pose, anchor_type)`, which are verified by ST-CM before execution.

The framework includes five specialized agents, each responsible for a distinct stage of adaptive MR content placement:

- **Spatial Perception Agent** converts raw sensor inputs from the MR device, including camera poses and detected surfaces, into a structured representation of the physical environment to support geometry-aware reasoning.
- **User Context Agent** analyzes the project structure and user-provided semantic cues to infer intent, scene relationships, and adaptation requirements across different physical settings.
- **Retrieval-Augmented Generation Agent** resolves under-specified content and layout decisions by querying project-level or external knowledge sources and producing high-confidence contextual constraints.
- **Interaction Agent** determines which virtual objects should support user interaction and assigns suitable behaviors by combining spatial context, object semantics, and available interaction primitives.
- **Human Interface Agent** synthesizes spatial constraints, interaction specifications, and temporal ordering to generate the final executable MR layout, including object placement, anchoring decisions, and scene transitions.

Together, these agents separate high-level semantic reasoning from low-level geometric verification: agents propose intent-aligned constraints and interactions, while ST-CM enforces physical feasibility and temporal/scene consistency before emitting executable layout commands. This separation is essential for robust adaptation across diverse environments and multi-scene MR experiences.

3 EVALUATION

We evaluated results aggregated over **15 scenarios** (**5 projects** $\times$ **3** real-world indoor environments), each repeated twice.

Metric (physical feasibility). We measure physical placement quality using *placement error (%)* defined as the percentage of placed objects that violate at least one feasibility constraint under the sensed geometry: (i) *support violation* (no valid contact with a supporting surface), (ii) *collision/penetration* with reconstructed geometry or other objects, or (iii) *floating artifacts* (visibly suspended without support).

Quantitative results. Compared to a single-prompt LLM baseline, enabling SpatialLLM’s spatial memory and constraint verification reduces average placement error from 65.2% to 8.6%, indicating substantially fewer infeasible placements in complex scenes.

User study (semantic plausibility). A user study with 11 participants indicates that SpatialLLM’s placements are perceived as more coherent with everyday spatial expectations (e.g., objects placed on plausible surfaces rather than intersecting walls), and as more stable and better adapted to surrounding environments. Qualitative feedback highlighted the benefits of automatic adaptation and collision handling in cluttered spaces.

4 CHALLENGES AND FUTURE WORK

Although SpatialLLM demonstrates the potential of constraint-verified, LLM-driven spatial reasoning for adaptive MR content placement, several challenges remain. First, the current framework primarily targets single-user scenarios. Extending to multi-user settings will require shared spatial memory, coordination, and conflict resolution for collaborative interaction.

Second, while layouts adapt to an initial environment scan, real-time responses to dynamic changes (e.g., moving furniture, people entering the space) remain limited. Future work will explore incremental ST-CM updates and asynchronous execution to improve responsiveness.

Finally, the framework inherits challenges associated with LLMs, including non-deterministic outputs and model variability. SpatialLLM mitigates these effects through constraint validation and constrained re-generation, but model drift remains a concern. Future directions include confidence estimation for spatial decisions, domain-adapted models, and mixed-initiative interaction that allows users to refine or override automatic placements.

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